1.) (a) Approximate the function $\sqrt[3]{x}$ by a Taylor polynomial of degree 2 at $a = 8$.

(b) How accurate is this approximation when $7 \leq x \leq 9$?

2.) (a) What is the maximum error possible in using the approximation

$$\sin(x) \approx x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

where $-0.3 \leq x \leq 0.3$? Use this approximation to find $\sin 12^\circ$ correct to six decimal places.

(b) For what values of x is this approximation accurate to within 0.00005?

3.) Use power series to estimate the integral $\int_0^{0.5} x^2 e^{-x^2}\, dx$ with an absolute error than 0.001.